

PENANG INTL, Malaysia

WMO: 486010

Lat: 5.2956N

Lon: 100.2680E

Elev: 3

StdP: 101.28

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	23.2	23.8	18.2	13.1	27.8	19.1	13.9	27.7	6.5	28.3	5.7	29.0	1.5	350	0.356

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	6.1	33.2	26.3	33.0	26.3	32.6	26.3	28.0	31.6	27.7	31.2	27.3	30.9	3.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.1	22.8	30.7	26.7	22.2	30.2	26.2	21.6	29.7	90.2	31.5	88.0	31.7	86.8	31.4	31.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.2	5.4	5.0	DB	21.5	34.8	1.8	1.6	20.2	36.0	19.2	36.9	18.2	37.8	16.8	39.0	(4)
(5)			WB	19.3	28.8	1.0	0.9	18.6	29.4	18.0	29.9	17.5	30.4	16.7	31.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	28.2	28.2	28.5	28.7	28.7	28.6	28.3	28.0	27.8	27.7	27.7	27.8	(6)	
(7)		DBStd	1.03	0.87	0.94	1.06	0.93	0.95	0.98	1.00	1.01	0.97	0.90	0.81	0.91	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	6648	563	519	580	561	579	557	566	558	533	547	531	553	(10)
(11)		CDD18.3	3606	305	286	322	311	321	307	308	299	283	289	281	295	(11)
(12)		CDH23.3	40016	3296	3261	3774	3669	3825	3617	3519	3289	2986	2924	2816	3041	(12)
(13)		CDH26.7	14687	1178	1291	1548	1477	1527	1430	1316	1163	987	916	880	976	(13)
(14)	Wind	WSAvg	1.9	2.4	2.4	2.0	1.8	1.7	1.8	1.8	1.7	1.7	1.8	2.3	(14)	
(15)	Precipitation	PrecAvg	2385	76	81	149	206	216	169	192	249	333	359	237	120	(15)
(16)		PrecMax	3253	372	250	402	389	467	396	361	591	629	869	517	381	(16)
(17)		PrecMin	1516	0	0	2	17	64	22	30	105	128	99	45	5	(17)
(18)		PrecStd	347	66	61	90	87	102	83	79	105	118	141	112	77	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.2	34.0	34.2	33.8	33.2	33.2	33.1	33.0	32.6	32.8	32.2	32.8	(19)
(20)			MCWB	24.8	25.5	25.8	27.3	27.1	27.0	26.6	26.5	26.4	26.5	26.2	25.6	(20)
(21)		2%	DB	32.2	33.1	33.2	33.0	32.9	32.8	32.2	32.1	32.0	31.9	31.8	32.0	(21)
(22)			MCWB	25.0	25.4	26.1	27.0	27.0	26.9	26.5	26.3	26.3	26.4	26.2	25.5	(22)
(23)		5%	DB	31.9	32.3	32.8	32.2	32.1	32.1	31.9	31.7	31.2	31.1	31.1	31.2	(23)
(24)			MCWB	25.1	25.4	26.1	26.8	26.9	26.6	26.5	26.3	26.3	26.3	26.3	25.5	(24)
(25)		10%	DB	31.1	31.9	32.0	31.8	31.8	31.5	31.1	31.0	30.8	30.5	30.6	30.8	(25)
(26)			MCWB	25.1	25.4	26.2	26.8	26.8	26.6	26.4	26.3	26.2	26.1	26.1	25.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.1	27.6	28.3	28.3	28.2	27.7	27.6	27.5	27.6	27.6	27.3	(27)	
(28)			MCDWB	30.9	31.9	32.2	32.3	31.8	31.7	31.3	31.3	31.3	31.2	30.8	30.9	(28)
(29)		2%	WB	26.5	26.9	27.5	27.8	27.8	27.7	27.3	27.1	27.0	27.0	27.0	26.7	(29)
(30)			MCDWB	30.3	31.0	31.5	31.5	31.2	31.0	30.8	30.7	30.5	30.6	30.5	30.3	(30)
(31)		5%	WB	26.2	26.5	27.0	27.4	27.5	27.3	26.9	26.7	26.6	26.6	26.6	26.2	(31)
(32)			MCDWB	29.9	30.5	30.9	31.0	30.8	30.6	30.4	30.2	29.9	30.0	29.9	29.7	(32)
(33)		10%	WB	25.8	26.2	26.6	27.0	27.1	26.9	26.6	26.5	26.3	26.3	26.2	25.9	(33)
(34)			MCDWB	29.4	30.0	30.4	30.5	30.4	30.3	29.9	29.8	29.4	29.4	29.4	29.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.7	7.1	6.9	6.4	6.1	6.2	6.2	6.1	6.1	6.1	6.1	6.2	(35)
(36)			MCDBR	7.2	7.6	7.6	6.9	6.7	6.7	6.6	6.7	6.7	6.7	6.8	6.8	(36)
(37)			MCWBR	3.5	3.5	3.2	3.0	2.8	2.8	2.8	2.9	2.9	2.9	3.0	3.2	(37)
(38)		5% WB	MCDWB	6.7	7.0	6.9	6.4	6.2	6.2	6.2	6.3	6.3	6.3	6.4	6.4	(38)
(39)			MCWBR	3.4	3.4	3.2	3.1	2.8	2.8	2.9	3.0	2.9	3.0	3.0	3.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.445	0.479	0.489	0.493	0.470	0.486	0.487	0.489	0.482	0.485	0.455	0.437	(40)	
(41)		taud	2.370	2.242	2.245	2.238	2.322	2.290	2.271	2.254	2.275	2.265	2.357	2.410	(41)	
(42)		Ebn at Noon	877	853	843	825	825	800	804	818	837	837	861	877	(42)	
(43)		Edh at Noon	125	145	146	144	129	131	134	140	139	140	126	119	(43)	
(44)	All-Sky Solar Radiation	RadAvg	5.10	5.81	5.75	5.56	5.11	5.05	4.94	4.95	4.76	4.69	4.65	4.62	(44)	
(45)		RadStd	0.36	0.34	0.43	0.20	0.24	0.23	0.22	0.24	0.29	0.39	0.24	0.37	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.33	+0.14	N/A	+0.40	+0.38	+0.27	N/A	N/A	+121	+121	(46)
(47)	Regional (10 neighbors)	+0.30	+0.18	N/A	+0.45	+0.36	+0.34	N/A	N/A	+121	+121	(47)

Nomenclature: See separate page